

Maximilian Lombardo

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EXPERIENCE

Biohub (Chan Zuckerberg Initiative)

New York, NY / Remote

Senior Applied Scientist, Product - AI Research and Cell Science

Nov 2020 – Present

- Partnered with product leadership to shape roadmap inputs, specifications, and execution plans for **CELLxGENE**, a scientific data platform used by **10K+ researchers**. Translated scientific and user requirements into product workstreams across visualization, data discovery, API access, embeddings, and AI model workflows.
- Coordinated releases of **9 biological foundation models** to the Virtual Cells Platform, spanning internal model launches, external model submissions, technical documentation, model evaluation, and cross-functional execution across AI Research, Engineering, Product, and Communications.
- Embedded with AI research and engineering teams during biological foundation model training to identify early applications, evaluate emerging capabilities, and translate progress into product use cases, demos, and roadmap recommendations.
- Built a RAG pipeline to mine biological foundation model literature, extract use-case patterns, identify adjacent tools, and discover workflow clusters across modalities; used findings to inform application discovery and product strategy.
- Conducted biosafety filter analysis for protein language model applications, using SAE feature analysis to identify dangerous-protein query risks and inform production safeguards for model-enabled biological workflows.

Kallyope

New York, NY / Remote

Computational Scientist - ML/AI

Sep 2017 – Nov 2020

- Led computational analysis generating a single-cell atlas of a novel tissue, partnering with wet-lab scientists to characterize cell types via graph-based clustering. Delivered insights securing a **\$2M milestone** with Novo Nordisk.
- Engineered ML pipelines, supporting company growth from **Series A (pre-clinical)** to **Series D**, contributing to identification of lead compounds for the **first human clinical trials**. Co-authored publication in *Cell Metabolism*.
- Co-developed probabilistic ML models to map neuronal connections, integrating viral tracing with single-cell sequencing data to decode circuit architecture.
- Presented research progress to the Scientific Advisory Board (including **three Nobel Laureates**), synthesizing complex computational results into strategic updates.

IBM Corporation

Amsterdam, NL

Graduate Machine Learning Research Intern - Center For Advanced Studies

Oct 2016 – Aug 2017

- Implemented and benchmarked machine learning frameworks to identify disease-causing features from multi-omics data.
- Evaluated dimensionality reduction techniques (subspace projections) for visualizing single-cell patient data.

Columbia University Medical Center

New York, NY

Senior Technician - Experimental Therapeutics for Lymphoma

Jul 2013 – Aug 2015

- Uncovered drug resistance via RNA-seq, resulting in **four co-authored publications** (Blood, Clinical Cancer Research).

EDUCATION

University of Amsterdam

Amsterdam, NL

Master of Science in Computational Science

Class of 2018

Columbia University in the City of New York

New York, NY

Master of Arts in Biotechnology

Class of 2014

Siena College

Albany, NY

Bachelor of Science in Biology, Cum Laude

Class of 2012

SKILLS

Languages & AI: Python, PyTorch, SQL, JavaScript, React, RAG, AI Agents, Model Evals, LLMs

Product & Tools: Product Strategy, GTM, Claude Code, AWS, Figma, Marimo, Remotion, Git, User Research

VENTURE PROJECTS

Stealth Biotech Venture Concept (Founder): Defined product vision for an AI-native "Operating System for Biology". Recruited early collaborators and pitched the comprehensive SaaS roadmap to venture capital firms.

Bay Bridge Bio Venture Exploration Project: Developed venture concept and MVP for a SaaS product in the regenerative medicine space, solicited KOL feedback, and pitched to Andreessen Horowitz, Leaps by Bayer, and J&J Innovation.